IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

PROBABILITY AND STOCHASTIC PROCESSES

Tuesday, 28 April 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for students

Each of the four questions has 25 marks.

The Questions

1. Random variables.

   a) Alice and Bob agree to meet in h-bar after their Friday lectures. They arrive at times that are independent and uniformly distributed between 5:00pm and 5:40pm. Each is prepared to wait s minutes before leaving. Find a minimal s such that the probability that they meet is at least 3/4.

   [15]

   b) The random variables X_i (i = 1, 2, …, n) are i.i.d. and uniform in the interval [0, 2]. Find the probability density function $f(y)$ for $Y = \max\{ X_i \}$.

   [10]

2. Estimation.

a) The random variable X has the Gamma distribution

$$f(x) = \frac{\beta^{\alpha} x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}, \quad x > 0, \alpha = 10, \beta > 0.$$

We observe the i.i.d. samples x_i = 3.9, 7.3, 5.4, 7.2, 6.2. Find the maximum-likelihood estimate of parameter β.

[10]

b) Consider the Rayleigh fading channel in wireless communications, where the channel gains $Y(n)$ have autocorrelation function

$$R_Y(n) = J_0(2\pi f_d n)$$

where J_0 denotes the zeroth-order Bessel function of the first kind, and f_d represents the normalized Doppler frequency shift. Suppose we wish to predict $Y(n+1)$ from $Y(n)$, $Y(n-1)$, …, $Y(1)$ using the linear MMSE estimator

$$Y(n+1) = \sum_{i=1}^{n} c_i Y(i)$$

Let $f_d = 0.3$ and given the following values of J_0 for $f_d = 0.3$:

$$J_0(2\pi f_d n) = \begin{cases} 1 & n = 0 \\ 0.291 & n = 1 \\ -0.402 & n = 2 \end{cases}$$

i) Compute the coefficient and mean-square error of the first-order linear MMSE estimator, i.e., $n = 1$. [6]

ii) Compute the coefficients and mean-square error of the second-order linear MMSE estimator, i.e., $n = 2$. [9]

3. Random processes.

a) Consider the random process $X(n) = A\cos(n\lambda) + B\sin(n\lambda)$ for parameter $\lambda > 0$, where A and B are uncorrelated random variables with zero mean and unit variance.

i) Find the mean and autocorrelation function of $X(n)$. [4]

ii) Is $\{X(n)\}$ wide-sense stationary? Justify your answer. [2]

iii) Is $\{X(n)\}$ strict-sense stationary? Justify your answer. [4]

b) The number of tutees $N(t)$ arriving at a professor's office over the time interval $[0, t)$ can be modelled by a Poisson process $\{N(t), t \geqslant 0\}$. On average, there is a new tutee arriving after every 5 minutes, i.e., the intensity of the process is equal to $\lambda = 0.2$. The professor will not start the tutorial until at least 3 tutees are in the office.

i) Find the expected waiting time until the tutorial starts.

[3]

ii) What is the probability that the tutorial does not start in the first 15 minutes?

[6]

iii) What is the probability that at least one tutee arrives in the first 5 minutes while at most two tutees arrive in the second 5 minutes?

[6]

4. Markov chains and martingales.

a) Calculate the stationary distribution for a Markov chain with state space $E = \{1, 2, 3, \ldots, M\}$, whose transitional probability matrix is

$$\begin{pmatrix} 2/3 & 1/3 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 1/3 & 1/3 & 1/3 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1/3 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1/3 & 2/3 \end{pmatrix}$$

[8]

b) Consider a Markov chain $X_0, X_1, X_2, \ldots$ with state space $E = \{0, 1, 2, 3, \ldots, N\}$. The transitional probability from state j to state k is given by the binomial distribution

$$p_{jk} = \binom{N}{k}\left(\frac{j}{N}\right)^k\left(1 - \frac{j}{N}\right)^{N-k}$$

i) Calculate $E[X_{n+1}|X_0, \ldots, X_n]$ and determine whether the Markov chain $\{X_n\}$ is a martingale or not. [8]

ii) Define another Markov chain $Y_n = X_n(N - X_n)$. Calculate $E[Y_{n+1}|X_0, \ldots, X_n]$ and determine whether $\{Y_n\}$ is a martingale with respect to $\{X_n\}$ or not. [9]